

(1.44-inch) coloured TFT screen display with SPI serial bus was used during testing. It is compatible with Nokia 5110, and is available on www.aliexpress.com

SDS011. Nova dust sensor SDS011 is one of the best



Fig. 3: ESP32 NodeMCU



Fig. 4: SDS011

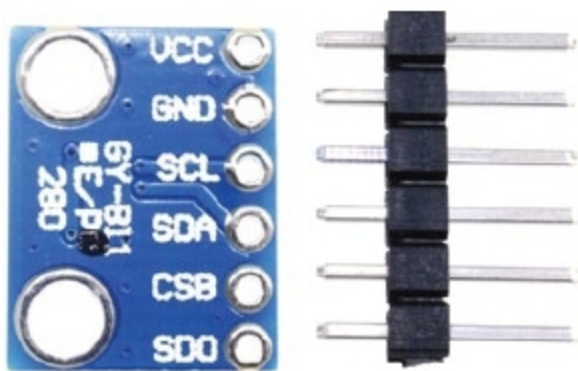


Fig. 5: BME280 module with male Berg connector

particulate sensors in terms of size, accuracy and price (available for US\$ 20 on www.aliexpress.com). It has a UART interface and measurement resolution of $0.3\mu\text{g}/\text{m}^3$. Operating voltage is 5V. SDS011 used in this project is shown in Fig. 4.

BME280. For reading temperature, pressure and relative humidity, an inexpensive BME280 module is used. BME280 used in this project is shown in Fig. 5.

2N2219. Transistor 2N2219 (T1) is used as a switch for SDS011. Its collector is connected to GND pin of SDS011, and its emitter to circuit ground. When pin 15 of ESP32 is logic high, T1 conducts and SDS011 gets ground connection through T1, otherwise SDS011 is off.

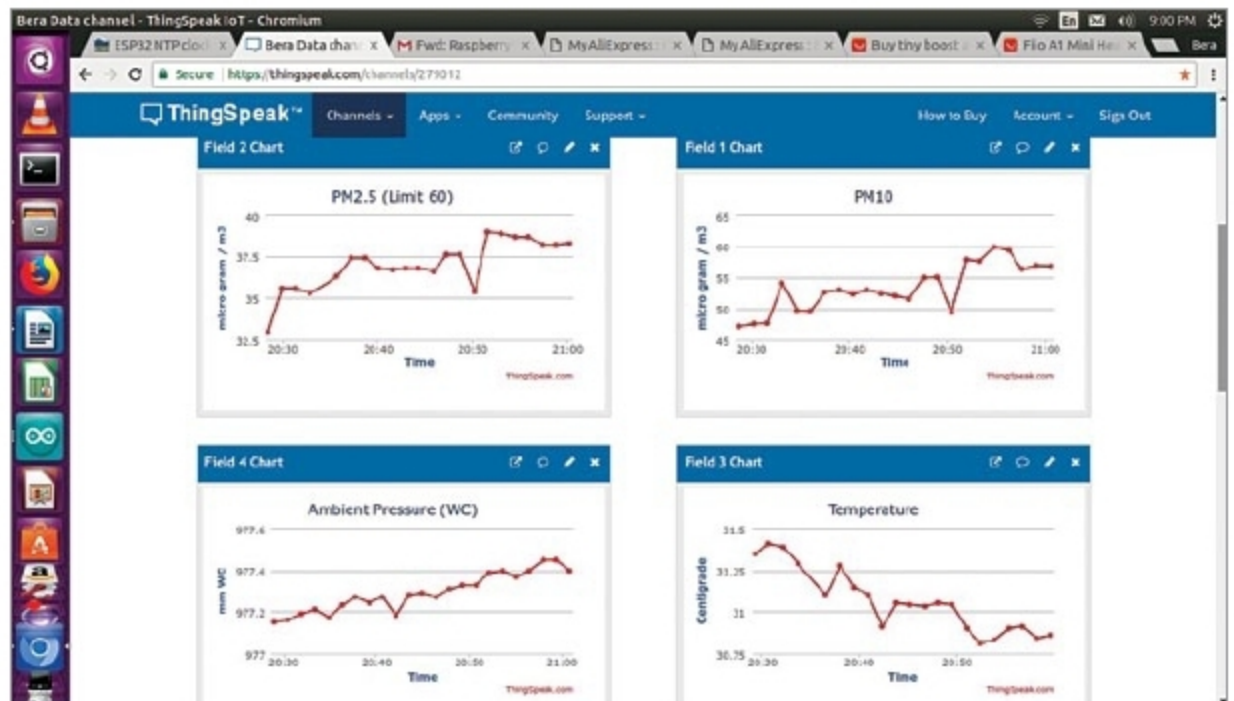


Fig. 6: Various parameters being displayed on ThingSpeak.com

ThingSpeak IoT platform

The project is based on ThingSpeak cloud computing. ThingSpeak is an open source IoT application and API to store and retrieve data from things using HTTP protocol over the Internet via LAN. It enables the creation of sensor-logging applications, location-tracking applications and a social network of things with status updates. This means, if you send data from the sensors to ThingSpeak at regular intervals, it will create, store and display data in a trend automatically. It has Maths modelling inbuilt in the form of MATLAB documentation, which is free to use.

Account and channel set up. Open an account and a channel on www.thingspeak.com. For this, you need to have a valid email account. The site will send a confirmation email. Click on the link sent in the email to validate your account and create a channel.

From here on, you can create as many channels as you want. The moment you create a channel, you get three ThingSpeak identities: channel ID, write API key and read API key. Note these down, as you will require these in the source program later. Feed up to eight sensor data per channel, such as PM2.5, PM10, temperature, relative humidity, altitude and pressure.

Eight data per channel. Each channel can take eight data signals from different devices. This means, using ThingSpeak API, you can upload eight data per channel, which are eventually gathered, logged and put

into trend data by ThingSpeak, like the following:

<https://thingspeak.com/channels/279012>

Software

The software (environment_thingSpeak.ino) for the project is written in Arduino programming language. PM2.5 and PM10 have a relation with relative humidity (RH). Therefore an RH correction is included in the program code (or sketch) to fine tune the readings.

SDS011 on ESP32. ESP32 works on Arduino IDE. Since Arduino is an advanced technology, there is a library available for almost all sensors. So, for SDS011 there is a ready-made library available that uses SoftwareSerial library and works out-of-the-box for Arduino Uno.

ESP32 does not work with SoftwareSerial. It has three UART ports and uses HardwareSerial for connecting UART devices. Existing library and command are used with hardwareSerial to get PM data from the sensors. These commands are placed in processdata.h file, which must be kept in the same folder as Arduino sketch and compiled together with the main sketch.

Sleep mode of SDS011. As per specifications, SDS011 has a sleep current of 2mA. The sensor works on sleep mode infrequently. Sleep commands are also placed in processdata.h file for you to try. However, GPIO13 (pin 15) is used to switch on the npn transistor (2N2219) to keep SDS011 on during active stage, before it goes into power-off mode.